

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

May 2012

EE-204 Electrical Power System-I

Date: 16.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What is string efficiency? Explain the different methods of improving the string efficiency. [07]

Q - 2 (a) Explain the A.C. power supply scheme with neat sketch. [07]

- (b) A string of four suspension insulators is connected across a 285 kV line. The self capacitance of each unit is equal to 5 times pin to earth capacitance. Calculate: [07]
- (i) The potential difference across each unit (ii) the string efficiency.

OR

Q - 2 (a) Explain with well labeled diagram the construction of Three core cable and explain function of each part. [06]

(b) Explain the causes and disadvantages of Low power factor. [04]

(c) A single phase motor connected to 400 V, 50 Hz supply takes 31.7 A at a power factor of 0.7 lagging. Calculate the capacitance required in parallel with the motor to raise the power factor to 0.9 lagging. [04]

Q - 3 (a) Define the following terms. [04]

- (1) Load factor
- (2) Plant capacity factor
- (3) Plant use factor
- (4) Load duration curve

(b) Explain the Sinking fund method of determining the depreciation. [05]

(c) A generating Station has the following daily load cycle: [05]

Time(Hours)	0-6	6-10	10-12	12-16	16-20	20-24
Load(MW)	40	50	60	50	70	40

Draw the load curve and find (i) Maximum demand (ii) Units generated per day

(iii) Average load (iv) Load factor.

OR

- Q - 3 (a) What is grading of cable? Explain the capacitance grading with neat sketch. [06]
- (b) What are the advantages of S.L. type cable over H type cable? Explain the H type cable with figure. [04]
- (c) A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 cm^2 [04] and internal diameter of sheath is 2.18 cm. the permittivity of the dielectric used in the cable is 3.5. find (i) the maximum electrostatic stress in the cable (ii) minimum electrostatic stress in the cable (iii) capacitance of the cable per km length (iv) charging current.

SECTION – II

- Q - 4 (a) What do you mean by Transposition of line? Derive the equation of capacitance for 3- Phase transposed line with Unsymmetrical spacing. [07]
- Q - 5 (a) Write technical note on Bundle conductor. [05]
- (b) Explain Self G.M.D. and Mutual G.M.D. [02]
- (c) Find the inductance per phase per km of double circuit 3-phase line as shown in following figure. The conductors are transposed and are of radius 0.75 cm each. The phase sequence is ABC. [07]

OR

- Q - 5 (a) Derive the equation of Flux linkages due to single current carrying conductor. [07] Consider two cases: (internal and external flux).
- (b) A 3-phase overhead transmission line has a series impedance of $(10 + j30)$ per phase. [07] For receiving end and sending end voltages of 132 kV and 140 kV respectively. Draw receiving end power circle diagram. Determine (i) The maximum real power delivered by the line and the load power factor under that condition. (ii) The capacity of shunt compensation equipment for supplying a load of 150 MVA at 0.8 p.f. lagging and the power angle under that condition.
- Q - 6 (a) Explain the End condenser method with vector diagram and give the limitation of this method. [07]
- (b) A 3-phase, 50 Hz transmission line 100 km long delivers 20 MW at 0.9 p.f. lagging [07] and at 110 kV. The resistance and reactance of the line per phase per km are 0.2Ω and 0.4Ω respectively, While capacitance admittance is 2.5×10^{-6} siemen/km/phase. Calculate: (i) Sending end voltage and current (ii) Transmission efficiency. Use Nominal T method.

OR

Q - 6 (a) Draw and explain π Model with its vector diagram. [07]

(b) Find the following for a single circuit transmission line delivering a load of 50 MVA [07]

at 110 kV and p.f. 0.8 lagging

(i) Sending end voltage (ii) Sending end current (iii) Sending end power

(iv) Transmission efficiency

Given $A=D=0.98 \angle 3^\circ$ ohm ; $B=110 \angle 5^\circ$ ohm; $C=0.005 \angle 80^\circ$ siemen.


